<u>EXPERIMENT 7</u>

Anomaly Detection using Unsupervised Learning

1. Problem Definition

The objective of this experiment is to detect **anomalies (outliers)** in a dataset using unsupervised learning algorithms.

Anomalies represent rare or unusual patterns such as:

- Fraudulent transactions
- Network intrusions
- Abnormal behavior

The problem is defined as:

- **1 → Anomaly (Outlier)**
- **0 → Normal Observation**

Unlike classification problems, anomaly detection is typically **unsupervised**, meaning:

- Models are trained without labeled outputs
- Labels are only used later for evaluation

2. Data Understanding

The dataset consists of **500 observations** with numerical features.

Extremely **imbalanced dataset (~6% anomalies)**

Class Distribution:

- Normal (0) - **470 samples**
- Anomaly (1) - **30 samples**

3. Data Preprocessing

The following steps were applied:

1. Missing Value Handling

- Removed null values

2. Feature Scaling

- Applied **StandardScaler**

4. Models Used

1. Isolation Forest

- Randomly partitions data
- Anomalies are easier to isolate - shorter paths
- Works well for high-dimensional data

2. Local Outlier Factor (LOF)

- Compares local density of a point with neighbors
- Points with lower density = anomalies

3. One-Class SVM

- Learns boundary around normal data
- Anything outside - anomaly

5. Evaluation and Prediction

Even though this is unsupervised, evaluation is done using true labels.

Metrics used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Model Performance Table

Model	Accuracy	Precision	Recall	F1-Score
Isolation Forest	0.96	0.60	1.00	0.75
Local Outlier Factor	0.844	0.02	0.0333	0.025
One-Class SVM	0.904	0.32	0.5333	0.40

Confusion Matrix Analysis

Isolation Forest

[[450 20]
 [0 30]]

- TN = 450 : correct normal
- FP = 20 : false alarms
- FN = 0 : **no anomalies missed**
- TP = 30 : all anomalies detected

Local Outlier Factor (LOF)

[[421 49]
 [29 1]]

- Detects **only 1 anomaly out of 30**
- Completely useless in real-world

One-Class SVM

[[436 34]
 [14 16]]

- Detects some anomalies (16/30)
- Misses many (14 anomalies)
- Moderate performance

6. Model Comparison Analysis

1. Accuracy is Misleading

- LOF has 84% accuracy but terrible recall
- Because anomalies are rare

2. Isolation Forest is the Best Model

- Recall = **1.0 (perfect)**
- F1-score = **0.75 (highest)**

3. LOF Completely Fails

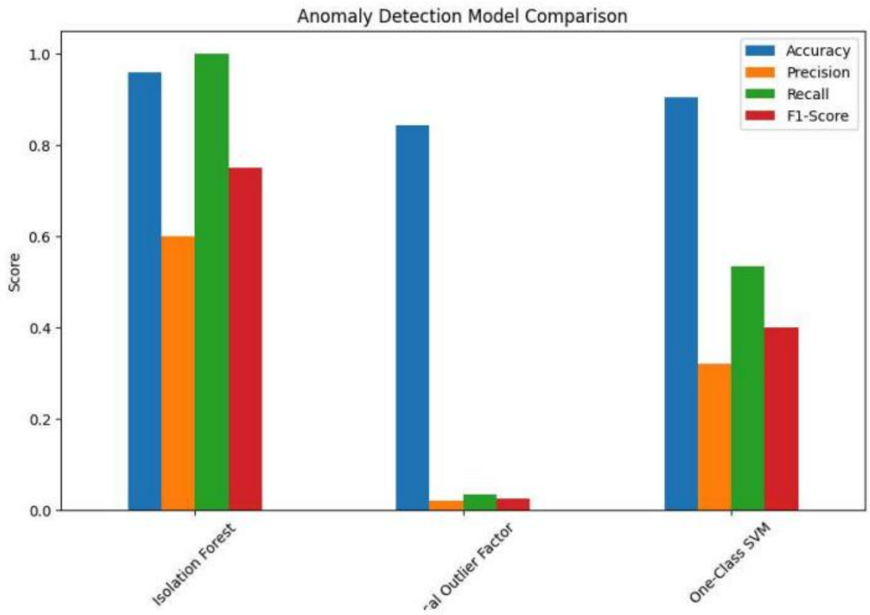
- Recall = **0.033**
- Detects almost no anomalies

4. One-Class SVM is Average

- Recall = **0.53**
- Better than LOF, worse than Isolation Forest

<u>**Visualization Analysis**</u>

- Isolation Forest dominates in recall and F1-score
- LOF almost flat - negligible performance
- One-Class SVM sits in the middle



Best Model Selection

Isolation Forest is the best model, because:

- Detects **100% anomalies (Recall = 1.0)**
- Highest F1-score
- Robust performance on imbalanced data